CariSurg MedTech Pathways Research Project

Preliminary Proposal

Project Title: AI-Assisted Emergency Department Triage Pilot at Mercer General Hospital

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Statement of the Problem:

Triage nurses in Mercer General Hospital's Emergency Department must rapidly assess patients during periods of high demand, increasing the risk of inconsistent prioritisation and delayed identification of high-acuity cases. Although recent studies suggest AI-assisted triage can improve patient prioritisation, most systems have not been validated using local emergency department data. This project proposes a 12-week pilot using de-identified Mercer ED records to evaluate whether AI-assisted acuity predictions align with current triage decisions and improve consistency in patient prioritisation

Previous Work:

The study is titled **"Use of Artificial Intelligence in Triage in Hospital Emergency Departments: A Scoping Review."** Emergency departments need faster and more accurate ways to triage patients. The authors reviewed studies from EMBASE, Ovid MEDLINE, and Web of Science to examine how AI and machine learning support ED triage. They found that machine learning improves triage prediction, disease identification, risk assessment, hospital admission prediction, and resource planning. The authors also noted that hospitals need more real-world testing across different settings because many studies used limited or retrospective data.[6]

The study **"AI-driven triage in emergency departments: A review of benefits, challenges, and future directions"** addresses overcrowding, limited resources, and inconsistent patient prioritisation in emergency departments. The authors reviewed peer-reviewed studies from 2015 to 2024 using PubMed, Scopus, IEEE Xplore, and Google Scholar. They found that AI-driven triage improves patient prioritisation, reduces waiting times, supports resource planning, and helps clinical staff manage high-pressure ED settings. The authors identified poor data quality, algorithmic bias, clinician trust, and ethical concerns as key barriers. They recommended stronger algorithms, better use of wearable technology, clinician training, and clear ethical guidelines before wider use in emergency care.[2]

The study **"TriageIntelli: AI-Assisted Multimodal Triage System for Health Centers"** focuses on emergency department overcrowding, limited resources, and the pressure on triage staff to prioritise patients quickly and accurately. The authors built and tested several AI models, including SVM, Random Forest, ANN, GBM, Linear Regression, XGBoost, and a stacking model, to predict patient triage levels using the Korean Triage and Acuity Scale. The stacking model produced the strongest result, with 80.05% accuracy, while SVM and GBM also performed well. The authors identified the single dataset as the main limitation because it reduces the model's use across hospitals with different patients, seasons, and clinical conditions.[5]

The study **"A Human-Centered Hybrid AI Framework for Optimizing Emergency Triage in Resource-Constrained Settings"** addresses the need for safer, faster emergency triage in healthcare facilities with limited staff, equipment, and clinical resources. The authors developed a human-centred AI system that uses patient vital signs, available resources, and clinician workload while allowing clinicians to review and override recommendations. The system achieved 90% accuracy, reduced missed high-acuity cases, shortened clinician decision-making time, and improved critical resource use. The authors identified the need for broader real-world validation and ongoing ethical and clinical supervision before full use.[1]

The study **"Application of artificial intelligence in triage in emergencies and disasters: a systematic review"** addresses the difficulty of triaging large numbers of injured patients during disasters and emergencies, where traditional manual methods slow decisions, limit patient tracking, and weaken resource allocation. The authors conducted a systematic review following PRISMA guidelines, searching PubMed, Scopus, Web of Science, Google Scholar, and article references up to September 2024. They reviewed 19 high-quality studies and found that AI and smart triage systems improved patient care through faster triage, real-time vital sign monitoring, better data sharing, and stronger resource management. The authors identified major limitations, including varied study designs, heavy use of simulated settings, English-only evidence, trust concerns, staff training needs, equipment shortages, and data privacy risks.[3]

"Use of Mobile Technologies to Streamline Pretriage Patient Flow in the Emergency Department: Observational Usability Study" examined whether smartphone-based self-registration could improve patient flow and reduce waiting times in emergency departments. The study used a mobile system that allowed patients to register using their own smartphones and QR codes before triage. Results showed that smartphone self-registration reduced pretriage waiting times, improved usability, and was preferred by triage nurses compared to kiosk-based registration. However, the study was conducted in only three hospitals in Shanghai and focused on usability and workflow improvements rather than clinical outcomes or patient safety impacts.[7]

"Development and Evaluation of Machine Learning Models for Emergency Severity Index Prediction in Febrile Tachycardic Adults"examined whether machine learning could improve the consistency of emergency department triage for patients with fever and tachycardia. The study compared Random Forest, Gradient Boosting Machine, and XGBoost models using routine triage data from 500 patients. XGBoost achieved the highest accuracy in predicting ESI levels and was integrated into the Thailand Triage Prediction System for real-time use. However, the study was limited to a single hospital, lacked external validation, and did not assess performance across different patient groups. [4]

Gap 1: Local validation against Mercer General's current triage decisions

Several studies report strong AI triage performance. AI and machine learning have been shown to improve triage prediction and risk assessment [6]. AI-assisted models have achieved approximately 80% accuracy in predicting triage levels [5], while human-centred AI frameworks have demonstrated improved identification of high-acuity patients [1]. However, these studies leave uncertainty regarding whether such systems would perform effectively in a different hospital environment. Single-dataset studies [5], reliance on retrospective evidence [6], and calls for broader real-world validation [1] suggest that local testing remains necessary. Mercer General cannot assume that a system validated elsewhere will safely identify high-acuity patients within its own emergency department. Local patient demographics, staffing patterns, documentation practices, and triage workflows may influence performance. Failure to identify high-acuity patients could contribute to under-triage and delayed care.Mercer could conduct a 12-week retrospective pilot using de-identified emergency department triage records. AI-assisted triage outputs could be compared with existing triage decisions and subsequent clinical outcomes. The primary outcome would be whether the AI system improves identification of high-acuity patients relative to current practice. This approach is feasible because it relies on existing data and does not alter live patient care.

Gap 2: Clinician trust and usability before live deployment

Clinician trust, ethical concerns, and training requirements have been identified as major barriers to AI triage adoption [2]. Similar concerns regarding trust, staff training, equipment availability, privacy, and implementation challenges have also been reported [3]. Human-centred frameworks that include clinician review and override capabilities further emphasise the importance of maintaining clinician involvement in decision-making [1].Although many studies focus on model accuracy, there is less evidence regarding whether emergency department clinicians understand, trust, and would realistically use AI recommendations during routine triage. This gap is important because even highly accurate systems may fail if clinicians do not trust or understand them. Poor trust may lead to ignored recommendations, over-reliance, or unsafe use. Emergency department leaders need evidence that AI supports clinical judgement rather than disrupting it.Mercer could conduct a 12-week shadow-mode pilot in which triage nurses review AI recommendations after completing their normal triage assessments. Nurses could evaluate whether recommendations were clear, useful, and clinically reasonable. The primary outcome would be clinician agreement with and perceived usefulness of AI-assisted triage recommendations.

Proposed Solution:

This project aims to build a small AI-assisted triage evaluation prototype for Mercer General Hospital's Emergency Department. The prototype will use de-identified past ED records. It will compare the estimated acuity level with existing triage decisions to assess consistency in patient prioritisation. The falsifiable target is that the prototype matches current triage decisions in at least 75% of reviewed cases and identifies high-acuity cases with at least 80% sensitivity. The project will include one simple prediction model, a basic results dashboard, and a final evaluation report.

References :

[1] E. Bhuvaneswari, K. D. V. Prasad, M. Ashraf, S. Jadhav, T. R. K. Rao, and T. Sudha Rani, “A human-centered hybrid AI framework for optimizing emergency triage in resource-constrained settings,” *Intell.-Based Med.*, vol. 12, p. 100311, Jan. 2025, doi: 10.1016/j.ibmed.2025.100311.
[2] A. Da’Costa, J. Teke, J. E. Origbo, A. Osonuga, E. Egbon, and D. B. Olawade, “AI-driven triage in emergency departments: A review of benefits, challenges, and future directions,” *Int. J. Med. Inf.*, vol. 197, p. 105838, May 2025, doi: 10.1016/j.ijmedinf.2025.105838.
[3] “Application of artificial intelligence in triage in emergencies and disasters: a systematic review | BMC Public Health | Springer Nature Link.” Accessed: Jun. 15, 2026. [Online]. Available: https://link.springer.com/article/10.1186/s12889-024-20447-3
[4] C. Wicha *et al.*, “Machine Learning-Based Model to Classify Emergency Severity Index Levels 1-3 in Febrile Patients With Tachycardia: Thailand Triage Prediction System,” *J. Clin. Med. Res.*, vol. 17, no. 12, pp. 676–687, Dec. 2025, doi: 10.14740/jocmr6371.
[5] Z. Araouchi and M. Adda, “TriageIntelli: AI-Assisted Multimodal Triage System for Health Centers,” *Procedia Comput. Sci.*, vol. 251, pp. 430–437, Jan. 2024, doi: 10.1016/j.procs.2024.11.130.
[6] “Use of Artificial Intelligence in Triage in Hospital Emergency Departments: A Scoping Review | Cureus.” Accessed: Jun. 15, 2026. [Online]. Available: https://www.cureus.com/articles/248163-use-of-artificial-intelligence-in-triage-in-hospital-emergency-departments-a-scoping-review#!/
[7] P. Wang, L. Yu, T. Li, L. Zhou, and X. Ma, “Use of Mobile Technologies to Streamline Pretriage Patient Flow in the Emergency Department: Observational Usability Study,” *JMIR MHealth UHealth*, vol. 12, p. e54642, Jun. 2024, doi: 10.2196/54642.